

1.2 Classwork: Introduction to Sequences

1. Consider the sequence: 2, 5, 8, 11, ...

(a) Write down the first term u_1 . [1]

(b) Is the sequence arithmetic, geometric, or neither? [1]

(c) Find the common difference d . [1]

(d) Find the next two terms in the sequence. [1]

(e) Find a general expression for u_n , the n^{th} term. [2]

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2. Consider the sequence: 4, 12, 36, 108, ...

(a) Write down the first term u_1 . [1]

(b) Is the sequence arithmetic, geometric, or neither? [1]

(c) Find the common ratio r . [1]

(d) Find the next two terms in the sequence. [1]

(e) Find a general expression for u_n , the n^{th} term. [2]

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3. An arithmetic sequence has first term $u_1 = 5$ and common difference $d = 4$.

(a) Write down the first four terms of the sequence. [2]

(b) Find u_{10} , the tenth term. [2]

(c) Find S_{10} , the sum of the first 10 terms. [2]

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4. A geometric sequence has first term $u_1 = 2$ and common ratio $r = \frac{3}{2}$.

(a) Write down the first four terms of the sequence. [2]

(b) Find u_7 , the seventh term. [2]

(c) Find S_7 , the sum of the first 7 terms. [2]

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